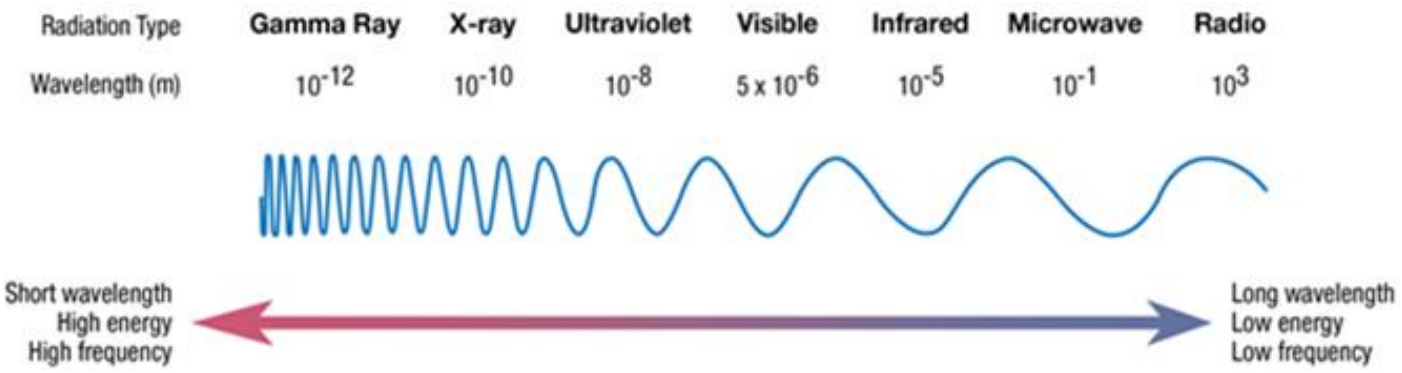
INSTRUMENTAL METHODS AND APPLICATIONS

Electromagnetic spectrum:

The electromagnetic spectrum includes all electromagnetic waves, from gamma rays to Radio waves. It's classified based on wavelength or frequency, with longer wavelengths at one end (radio waves) and shorter wavelengths at the other (gamma rays).

γ-rays < X-rays < U.V. rays < Visible rays < I.R. rays <Microwaves <Radio waves



1. **Gamma Rays**: Wavelength: < 0.01 nm. The highest-energy electromagnetic waves.
   Used in medical treatments- radiation therapy (cancer treatment), Nuclear industry - radioactive waste management.
2. **X-rays**: Wavelength: 0.01 nm to 10 nm.
   Used in medical imaging- diagnosing fractures and internal conditions, industrial inspection, security-scanning and scientific research.
3. **Ultraviolet (UV)**: Wavelength: 10 nm to 400 nm.
   Used in applications such as sterilization (killing bacteria and viruses), water treatment and health care-medical treatments.
4. **Visible Light**: Wavelength: 400 nm to 700 nm. The range of electromagnetic waves visible to the human eye, comprising different colors.
   Used in lighting fixtures and displays (LEDs, fluorescent lights), photography.
5. **Infrared (IR):** Wavelength: 700 nm to 1 mm. Associated with heat.
   Used in remote controls- electronic devices for wireless operation., night-vision equipment, infrared photography and thermal imaging.
6. **Microwaves**: Wavelength: 1 mm to 1 meter.
   Used in cooking (microwave ovens), radar, and certain types of wireless technology.
7. **Radio Waves**: Wavelength: Longest (1 mm to 100 km).
   Used for Telecommunications (radio, television, cell phones) broadcasting, and Navigation- GPS technology, radar etc.

UV Visible Laws of absorption: The phenomena of absorption of light by molecules can be governed by two laws. They are.,

1. *Lambert's law.* 2. *Beer's law.* 3. *Beer- Lamberts law.*

1. Lambert's law: When a beam of monochromatic light having intensity 'I' is made to fall perpendicularly on a homogeneous solution, the intensity of light is decreased (-dI) due to absorbance of light by the solution. If 'dx' is the small distance travelled by light in solution, the rate of decrease in intensity is represented by *'-dI/dx'.*

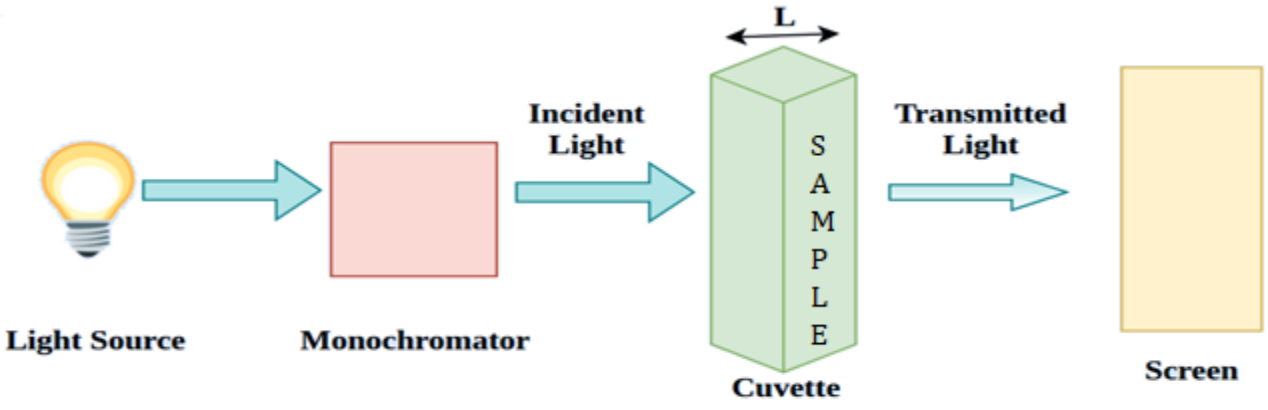
This rate of change (-dI/dx) in the intensity of light is directly proportional to the intensity (I) of incident radiation. This is called Lambert's law.

2. Beer's law: When a beam of monochromatic light enters perpendicularly into a dilute solution, the absorbance isdirectly proportional to the concentration of the solution. It is called Beer's law. Mathematically this can be written as,

$$log_{10}\frac{I_0}{I} = k'c$$

Where k' is a constant and 'c' is the concentration of the solution.

3. Beer-Lambert's law: It is the combined form of Beer's law and Lambert's law. According to this law, 'when a beam of monochromatic light is passed through a solution, the decrease in intensity of light with respect to the thicknessof solution is directly proportional to the intensity of incident radiation as well as to the concentration of the solution.



Beer Lambert Law: A = εLC

A= Absorbance ε = Molar extenction coefficient L = Path length C = Concentration of sample

ULTRAVIOLET (UV) SPECTROSCOPY

Ultraviolet (UV) spectroscopy, is an analytical technique that measures the absorption of UV light by a substance. It is commonly used to identify & quantify organic compounds and analyze their molecular structure within the 200nm to 400nm range.

Principle: When UV light passes through a sample, certain wavelengths are absorbed by the electrons in the molecules, which causes the electrons move from a ground state (Eg) to an excited state (Ee). The amount of light absorbed at specific wavelengths provides information about the electronic transitions in the molecule.

Then the difference in change of energy is $\Delta E = Ee - Eg = h\nu$

UV-Visible Spectrophotometer:

UV-Visible spectrophotometer is an analytical instrument**.** It is a combination of a spectrometer and a photometer.

i) Which is used to measure the absorption or transmission of ultraviolet (UV) and visible light by a sample or substance.

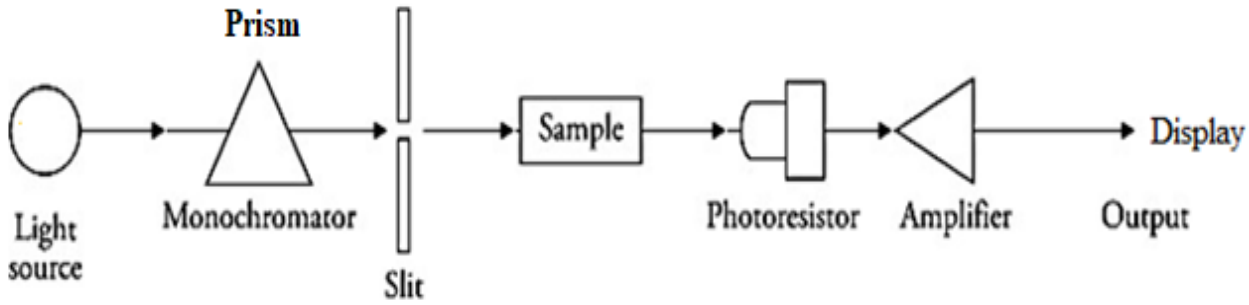
ii) It operates on the principle that molecules absorb or transmit % of light at specific wavelengths and providing information about their electronic structure.

Spectrophotometers are of two types, viz.,

1. Single beam spectrophotometer.
2. Double beam spectrophotometer.

Single beam spectrophotometer:

1. Light radiation is directed onto a prism or filter (monochromator).
2. A monochromatic beam of light is produced and passed through a slit and falls on the sample solution.
3. The sample absorbs certain frequencies of the incident light, and the remaining light goes through a photoresistor. The photoresistor converts the light into an electrical signal, which is sent to an amplifier.
4. The amplified signal is detected by output radiation detectors.
5. Reference solvent and sample solutions are manually changed in the sample holders to compare the percentage of absorbance.



THEORY OF ELECTRONIC SPECTROSCOPY: Electronic transitions in UV-spectra involve the movement of electrons between different electronic energy levels in molecules.

i) All molecules contain bonded electrons. Molecular orbitals are two types. They are bonding and anti-bonding orbitals. However non-bonding orbital contains electrons which did not participate in bonding.

ii) When molecules are irradiated with radiation of high frequency these bonded electrons can make excitations between molecular orbitals (σ, π, σ*, π*, n).

iii) All these electronic transitions follow selection rules. If the value of the molar extinction coefficient ε_{max} is more than 10^4, then those are called allowed transitions and less than 10^4, then those are called forbidden transitions.

Here are some common types of electronic transitions along with examples:

1. **<u>σ → σ* transitions</u>:** Involves the excitation of electrons from a sigma (σ) bond to the corresponding antibonding σ* orbital. This high frequency (~150nm) is in the UV region.

   Ex: Saturated hydrocarbons like *Methane, Ethane, Propane* etc. are detected in this region.

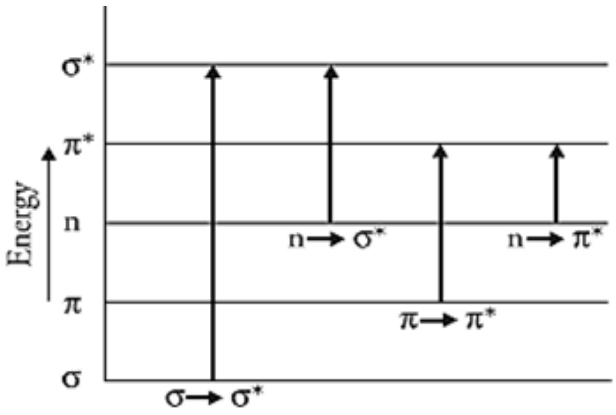
2. **<u>n → σ* transitions:</u>** Involves the excitation of non-bonding (lone pair) electrons (n-electrons) to the antibonding σ* orbital. At nearly 167–174nm frequency these transitions might occur.

   The hetero atoms like nitrogen, oxygen, Sulphur etc., are having unshared pair of electrons (non- bonding electrons) when bonded with a carbon atom in carbon compounds. Ex: *Ammonia (NH_3), Methyl alcohol,* etc., molecules give some spectral lines due to these transitions.

3. **<u>π → π* transitions</u>:** These type of transitions occurs in compounds having double and triple bonds. At nearly 170-190nm frequency these transitions might occur.

   Conjugated systems like ethene (C_2H_4), benzene (C_6H_6), etc. Molecules give some spectral lines due to these transitions.

4. **<u>n → π* transitions:</u>** These type of transitions occurs in compounds having hetero atom having unsaturated (double or triple) bonds like C=O (in *acetaldehyde-CH3CHO, Acetone-CH3-CO-CH3*), C=S etc. At nearly 280nm frequency these transitions might occur.



The relative energies required for the various transitions follow the order.

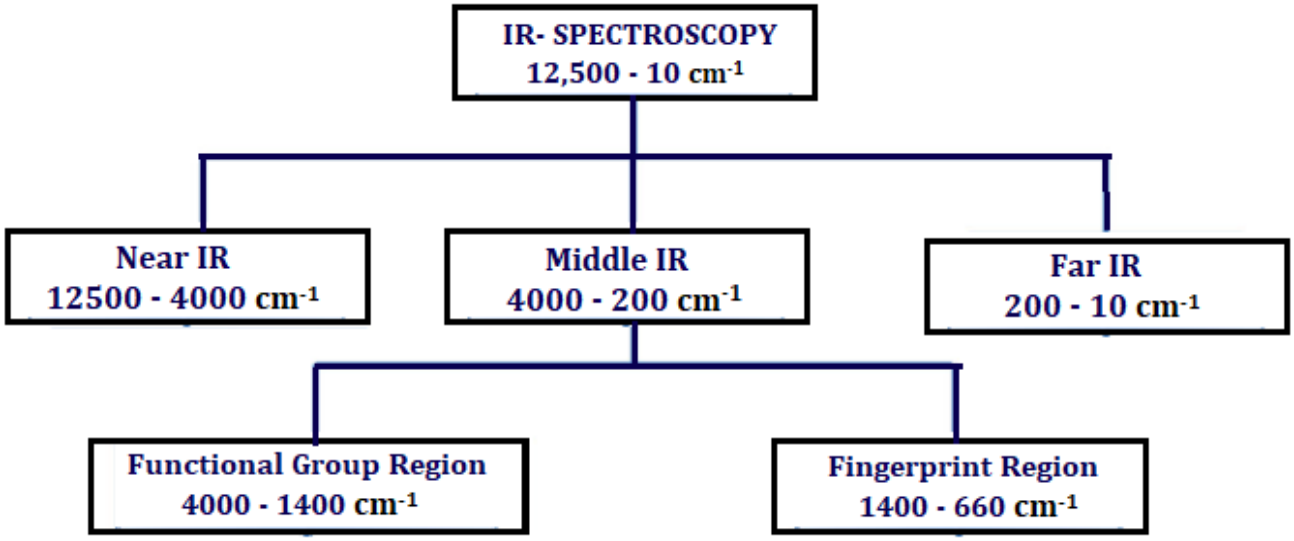
σ to σ* > n to σ* > π to π* > n to π*

Applications of UV visible spectroscopy:

1. Identification of unknown compounds and detect hydrogen bonds.
2. UV-V spectroscopy can be used to identify cis/trans isomers.
3. Used to find the presence of conjugation in molecules.
4. Used in analytical chemistry to determine purity of compounds.
5. Used in food and beverage sciences.
6. UV-V spectroscopy can be used in Pharmacology and Toxicology.
7. Used in biochemistry to measure structure of compounds and the concentration of DNA and RNA.

INFRARED (IR) SPECTROSCOPY

Infrared (IR) spectroscopy involves the interaction of molecules with infrared light. 12,500- 10 cm^{-1} (wave number) or 3.8 x 10^{14} – 30 x 10^{11} (Frequency range) is called an infrared region. When molecules absorb specific infrared frequencies, they undergo vibrational transitions, revealing information about their functional groups and structure. IR spectroscopy is valuable in identifying and characterizing organic/inorganic compounds in chemistry.



The infrared spectrum frequency range 4000-660 cm^{-1} is important for organic chemistry to detect the functional groups present in the molecules.

Different molecules absorb different IR frequencies. This nature is used to find the structure of compounds. *Fourier Transform Infrared (FTIR) spectroscopy* is the technique employed to find the structure of compounds in IR spectroscopy.

FTIR (Fourier Transform Infrared) SPECTROSCOPY:

i) Fourier Transform Infrared spectroscopy (FTIR) is a technique used to obtain an IR spectrum of materials.

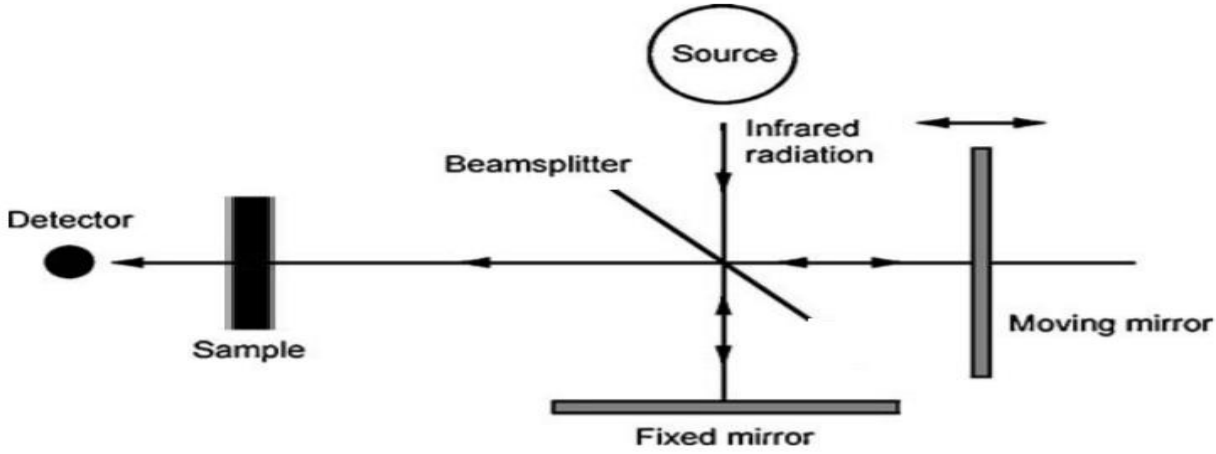
ii) The IR source sends an IR beam containing many frequencies of light at once. And it measures how much of that incident beam is absorbed by the sample. The incident beam first passes through the Michelson interferometer.

iii) Michelson interferometer consists of two types of mirrors. One mirror is fixed and the other is moving (can be moved by motor).

iv) By adjusting the position of 'moving mirror', light reflected by the moving mirror will undergo constructive interference with reflected light from the fixed mirror.

v) This infrared light passes through the sample. The vibrating groups in the sample absorb IR radiation of particular frequency and give resonance vibrations. Hence some amount of IR radiation is absorbed and the remaining comes out and falls on the detector.

vi) The total information was computed by computer and gives an IR spectrum.



Applications of IR spectroscopy:

1. Used to determine the structures of organic compounds.
2. Infrared spectroscopy can get information for the whole range of frequency simultaneously, within one second.
3. Used to find the nature of functional groups.
4. Used in analytical chemistry (food, beverages and forensic analysis).
5. Used in bio-chemistry (study of the structure and function of proteins, nucleic acids and other bio molecules)
6. In pharmaceutical industry, IR is used to ensure the quality of drugs and other medicinal products.

NUCLEAR MAGNETIC RESONANCE (NMR) SPECTROSCOPY

NMR is an abbreviation of *Nuclear Magnetic Resonance*. It is a type of spectroscopy technique based on the nuclear magnetic resonance of atoms of the sample being examined. This method allows us to determine the quality and purity of a sample, as well as the molecular structure of a compound.

Two common types of NMR spectroscopy are used to charecterise organic structure.

a) ^{1}H-NMR is used determine the type and no. of Hydrogen atoms in a molecule.

b) ^{13}C- NMR is used determine the type and no. of Carban atoms in the molecule.

The source of energy in NMR is radio waves which have long wave length and thus low energy and frequency.

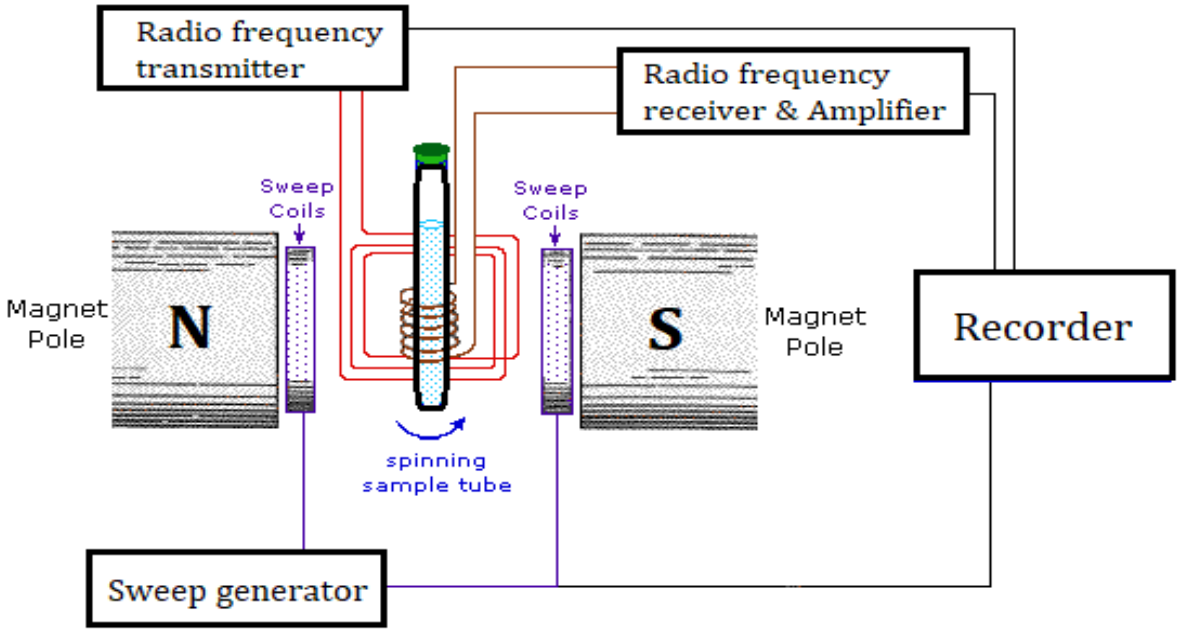
Principle of NMR Spectroscopy:

NMR spectroscopy operates on the principle that when atomic nuclei with spin are placed in a magnetic field, they can absorb and emit radiofrequency (region 3kHz-300 GHz) radiation. The resulting energy transitions provide valuable information about the molecular structure and environment of the sample under examination.

Instrumantation and Working of NMR- Spectroscopy:

i. The sample is dissolved in proton free solvent like $CDCl_3$ or CCl_4 and carefully placed within a magnetic field.

ii. The sample absorbs radio waves from radio frequency transmitter and the nuclei of the sample gets excited and generates specific NMR signals induced by nuclear magnetic resonance.



iii. Sweep generators modifies the strength of the magnetic field which is already applied with the help of sweep coils.

iv. These signals, which carry crucial information are carefully detected using highly sensitive radio frequency receivers.

v. The resonance frequency of an atom in a molecule changes because of the magnetic field within the molecule.

vi. This change in frequency enables a detailed analysis giving specific information about the molecule's functional groups and electronic structure.

vii. NMR spectroscopy serves as an absolute method for identifying monomolecular organic compounds, providing a wealth of information about a molecule's reaction state, structure, chemical environment, and dynamics.

Applications of NMR spectroscopy:

1. **Chemical Analysis:** Determination of molecular structures, elucidating chemical compositions, and identifying unknown compounds in chemistry, pharmaceuticals, and material sciences.
2. **Medical Diagnosis:** NMR imaging (MRI) is extensively used in medicine for non-invasive imaging to diagnose diseases and disorders, particularly in neurology, oncology, and musculoskeletal conditions.
3. **Protein Structure Determination:** Understanding the structure and dynamics of proteins, essential for drug design and development in the pharmaceutical industry.
4. **Metabolomics:** Studying metabolites in biological systems for understanding metabolic pathways, disease biomarkers, and evaluating drug efficacy.
5. **Quality Control in Industries:** Ensuring the purity and quality of food, beverages, polymers, and other industrial products.
6. **Environmental Studies:** Analyzing pollutants, identifying contaminants in soil and water samples, and studying environmental processes.
7. **Material Science:** Investigating the properties of materials, such as polymers and catalysts, to understand their structures and functionalities.
8. **Forensic Analysis:** Identifying trace evidence and analyzing compounds in forensic investigations.
9. **Petroleum Industry:** Assessing the composition of crude oil and its derivatives for refining processes.

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Unit-5 - TWO-mark Question and Answers

1. **What is the electromagnetic spectrum?**
   A: The electromagnetic spectrum is the range of all types of electromagnetic radiation, including γ-rays, X-rays, U.V. rays, Visible rays, I.R. rays, Microwaves and Radio waves
2. **State Beer-Lambert's law.**
   A: Beer-Lambert's law relates the absorbance of a substance to its concentration and the path length of the radiation, expressed as A = εcl, where A is absorbance, ε is molar absorptivity, c is concentration, and l is path length.
3. **What is UV-Visible spectroscopy?**
   A: UV-Visible spectroscopy is a technique that involves the study of electronic transitions in molecules using ultraviolet and visible light.
4. **Define electronic transitions in the context of UV-Visible spectroscopy.**
   A: Electronic transitions refer to the movement of electrons between different energy levels within a molecule, typically associated with absorption or emission of UV or visible light. (σ to σ* > n to σ* >π to π* >n to π*)
5. **Provide an example of an application of UV-Visible spectroscopy.**
   A: One application is the quantification of concentration in a solution, such as determining the concentration of a compound in a sample.
6. **Give an example of an application of FT-IR spectroscopy.**
   A: FT-IR spectroscopy is commonly used in the analysis of chemical composition, such as identifying functional groups in organic compounds.
7. **Briefly explain the principle behind Nuclear Magnetic Resonance (NMR).**
   A: NMR involves the interaction of atomic nuclei with an external magnetic field and radiofrequency radiation, resulting in the absorption of energy that can be detected and analyzed.
8. **What property of nuclei is crucial in NMR spectroscopy and mention it's key component?**
   A: The property of nuclear spin is crucial in NMR spectroscopy. its key component is the magnet.
9. **How does the strength of the magnetic field affect NMR spectroscopy?**
   A: The strength of the magnetic field influences the energy separation between nuclear spin states and affects the resolution and sensitivity of the NMR spectrum.
10. **Provide any two application of NMR spectroscopy.**
    A: **i)** NMR spectroscopy is used in medical imaging techniques such as Magnetic Resonance Imaging (MRI). **ii)** NMR provides information about the connectivity and environment of atoms in a molecule, aiding in the determination of its structure.

Department of Chemistry
Chemistry Question Bank (5 Marks)

Unit-1

1. Discuss postulates of Molecular Orbital Theory?
2. Illustrate MO diagram of O_2 and calculate bond order?
3. How do you calculate bond order of hetero diatomic molecule (CO) by using MOT?
4. Summarize pi-Molecular Orbital energy level diagram of 1, 3 Butadiene?
5. Describe pi Molecular Orbital energy level diagram of Benzene?
6. Explain Hydrogen bonding with examples?

Unit-2

7. Discuss Zone refining and Czochralski process with applications of semiconductors?
8. Discuss Classification of supercapacitors and its applications?
9. Explain types of superconductors with applications?
10. Illustrate preparation (Arc discharge & laser ablation methods) of nanomaterials.
11. Discuss properties and applications Graphenes nanoparticles?
12. Describe preparation fullerenes and its properties and applications?

Unit-3

13. Summaries electrochemical series and it's applications?
14. Discuss conductometric titration of strong acid Vs strong base?
15. Define potentiometric sensors and discuss it's working?
16. Illustrate zinc air battery with working and applications?
17. Define fuel cell and Illustrate H_2-O_2 and PEMF cell?
18. Describe construction and working of Li-ion/ Na -ion cell?

Unit-4

19. Distinguish difference between Chain polymerization and Step polymerization?
20. Distinguish difference between thermo plastics and thermo setting plastics.
21. Discuss Free radical & coordination polymerization with specific example?
22. Illustrate fabrication (Compression/ injection moulding) technique of plastics?
23. Describe Preparation, properties and applications of Buna-S/ Buna-N?
24. Define Conducting polymers? Explain conduction mechanism of polyacetylene with applications?
25. Discuss properties and applications of Bio-Degradable polymer of Poly Glycolic Acid (PGA)?

Unit - 5

26. Discuss electromagnetic spectrum with applications?
27. Discuss Beer-Lambert's law?
28. Illustrate UV – visible spectra instrumentation & applications of UV spectroscopy.
29. Discuss theory of electronic transitions in UV?
30. Demonstrate FT-IR instrumentation & applications of Infrared spectroscopy?
31. Demonstrate NMR instrumentation with applications?

Chemistry Question Bank (2 Marks)

Unit-I (Structure and bonding models)

1. What is a molecular orbital?
2. What are the types of molecular orbitals?
3. Calculate bond order of CO molecule?
4. Differentiate between Bonding and Antibonding Molecular Orbitals (any two)?
5. Why oxygen molecule is paramagnetic?
6. How electrons are filled in molecular orbitals?
7. Define Homonuclear Diatomic Molecules?
8. Define hydrogen bonding with an example?
9. What are the conditions for formation of hydrogen bonding?
10. Give two examples for intramolecular hydrogen bonding?
11. Explain why water has a high boiling point than methane and ethane?

Unit-II (Modern Engineering materials)

1. What are Semiconductors?
2. Give the principle involved in Zone refining process?
3. Give any two applications of semiconductors?
4. Define of superconductivity?
5. Define transition temperature (TC), give the transition temperature of Hg?
6. Define critical magnetic field (HC) with Meisner's effect?
7. Define supercapacitors?
8. What is Graphene and mention any two properties?
9. What are nanotubes and write any two applications?
10. What are fullerene and write any two applications?

Unit –III (Electrochemistry and Applications)

1. Define cell and give an example?
2. Define Battery?
3. What is an electrochemical cell?
4. What is the electrochemical series?
5. Define potentiometry in chemistry?
6. What is conductivity in the context of chemistry?
7. What are potentiometry sensors, and can you provide an example?
8. What is a primary cell, and can you name an example?
9. What defines secondary cells, and can you name examples?
10. What are fuel cells, and provide an example?

Unit-IV (Polymer Technology)

1. Define addition polymerization with examples.
2. What is condensation polymerization? Mention any two examples.
3. Which plastics are soft on heating and hard on cooling? Mention its examples.
4. What are thermosetting plastics? Write its examples?
5. Mention any two moulding(fabrication) methods.
6. Which catalyst is used in free radical addition polymerization?
7. Mention the monomers in BUNA-S.
8. Define conducting polymers with any two examples.
9. What are Biodegradable polymers with examples?
10. Define co-polymerization with examples.

Unit-5 (Instrumental Methods and Applications)

1. What is the electromagnetic spectrum?
2. State Beer-Lambert's law.
3. What is UV-Visible spectroscopy?
4. Define electronic transitions in the context of UV-Visible spectroscopy.
5. Provide an example of an application of UV-Visible spectroscopy.
6. Give an example of an application of FT-IR spectroscopy.
7. Briefly explain the principle behind Nuclear Magnetic Resonance (NMR)?
8. What property of nuclei is crucial in NMR spectroscopy and mention its key component?
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10. write any two applications of NMR spectroscopy.

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